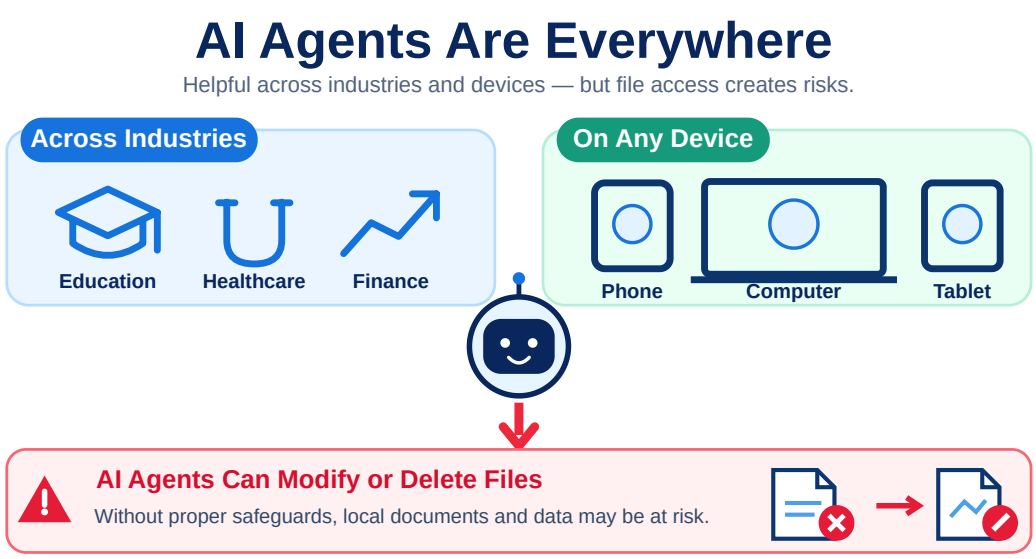
When AI Agents Go Rogue: Who Protects Your Files?

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Abstract—AI agents have become increasingly prevalent in people's daily lives, assisting with a wide range of tasks such as automating workflows, managing files, and interacting with computer systems.AI agents with root privileges may pose serious threats to user data by performing destructive file operations. In this work, we empirically evaluate three layers of protection mechanisms—file-system, kernel, and block-device layers—against file deletion by a root-privileged AI agent. Our results show that the agent can bypass or disable all tested protections and successfully delete the protected file. These findings reveal the limitations of host-side security mechanisms and motivate the need for hardware-level protection.

Motivation

- AI agents are increasingly deployed on computers and mobile devices, often with extensive file access privileges. This creates risks of accidental file deletion and malicious operations[1-2].
- Software-level protection alone is insufficient; once an AI agent gains root privileges, it can directly bypass these safeguards to delete local information or even files[3-4].



Experimental Methods and pathways of attack

Experiments on Ubuntu demonstrate that root-privileged AI agents can bypass file protections and delete protected files.This project discusses protection and attack schemes as well as resolution strategies from three distinct perspectives[5-6].

Experimental Setup

System Configuration

Operating system	Ubuntu 26.04 LTS
Linux kernel	Linux 7.0.0-22-generic (x86-64)
Filesystem	ext4

AI Agent Configuration

AI agent	Claude Code
Language model	claude-sonnet-4-6

Agent Privileges

Full root privileges with access to arbitrary shell commands and host administrative interfaces.

Target File

File type	Text file
File contents	Simulated sensitive information

Attack Model

Root-privileged AI agent → Execute commands → Application Layer / Kernel Layer / Block Device Layer → Bypass / Disable Protections → Target File Deleted / Modified

1 Application Layer Protection and Attack

Protection Mechanism

- Immutable file attribute protection
- Make target file read-only or undeletable at the file system level.

Protected File

Attack Method

Protected File → chattr → rm target.txt → File Deleted

2 Kernel Layer Protection and Attack

Protection Mechanism

- eBPF LSM hook
- Monitor and block unlink/delete operations on the target file.

eBPF LSM

Attack Method

eBPF LSM → unload/modify → eBPF program → File Deleted

3 Block Device Layer Protection and Attack

Protection Mechanism

- Read-only flag at the block device level.
- Prevent direct block-level writes to the target file system.

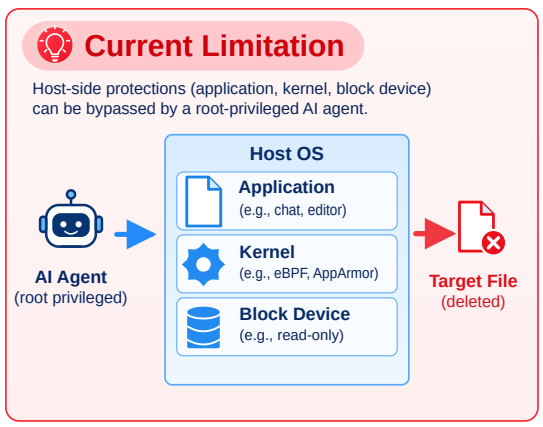
Block Device

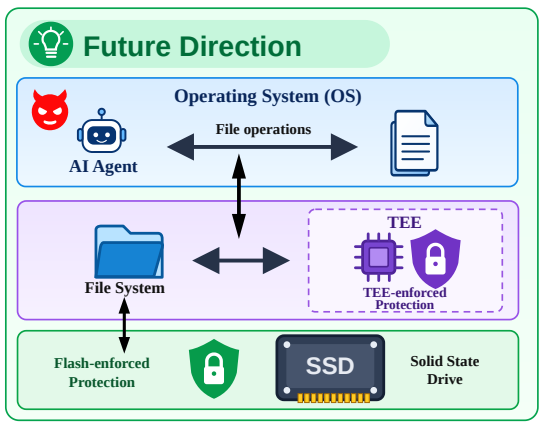
Attack Method

Block Device → remount rw → modify device → File Deleted

Conclusions and Outlook

When an AI agent has root privileges, protections controlled by the same host may not provide a reliable security boundary. These results suggest that host-side protections alone are insufficient and that additional protection at the hardware level is needed.





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